# Appendix A. Asymptotics

Bessel functions, Laplace's method and Stirling's formula, the Airy function, the partition function

## Contents

**1 Bessel functions** **2**

**2 Laplace's method; Stirling's formula** **3**

**3 The Airy function** **5**

**4 The partition function** **6**

## Overview

When a function cannot be computed explicitly, one studies its asymptotic behavior; complex analysis is of crucial help. Typically we study, for large $s$,

$$I(s) = \int_a^b e^{-s\Phi(x)}\,dx. \tag{1}$$

Three guiding principles:

(i) *Deformation of contour.* If $\Phi$ is complex-valued, the integrand oscillates rapidly for large $s$ and cancellations mask the behavior. When $\Phi$ is holomorphic one may change the contour so that $\Phi$ is essentially real on it, and then read off the behavior (illustrated with Bessel functions).

(ii) *Laplace's method.* If $\Phi$ is real and $s > 0$, the main contribution comes from near a minimum of $\Phi$, giving an expansion in terms of the quadratic behavior of $\Phi$ there (Stirling's formula, the Airy function).

(iii) *Generating functions.* The asymptotics of a sequence $F_n$ can be studied via $F_n = \int_\gamma F(e^{2\pi iz})e^{-2\pi inz}\,dz$, $\gamma$ a segment of unit length in the upper half-plane, a variant of (1); we obtain the asymptotics of the partition function $p(n)$.

## 1 Bessel functions

Bessel functions arise in problems with rotational symmetry (e.g. the Fourier transform of a radial function on $\mathbb{R}^d$ involves $J_{d/2-1}$). For $\nu > -1/2$,

$$J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)} \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{2}$$

$J_{-1/2}(s) = \lim_{\nu\to -1/2} J_\nu(s) = \sqrt{2/\pi s}\cos s$ and $J_{1/2}(s) = \sqrt{2/\pi s}\sin s$; $J_\nu$ is elementary only for half-integral $\nu$. These examples suggest:

**Theorem 1.1**

$$J_\nu(s) = \sqrt{\frac{2}{\pi s}}\cos\left(s - \frac{\pi\nu}{2} - \frac{\pi}{4}\right) + O(s^{-3/2}) \text{ as } s\to\infty.$$

It suffices to study

$$I(s) = \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{3}$$

Let $f(z) = e^{isz}(1-z^2)^{\nu-1/2}$ in the plane slit along $(-\infty,-1]\cup[1,\infty)$, with the branch positive on $(-1,1)$. For $s > 0$ fixed, Cauchy's theorem on the rectangle $\gamma_{\epsilon,R}$ with vertices $-1+\epsilon$, $1-\epsilon$, $1-\epsilon+iR$, $-1+\epsilon+iR$ (letting $\epsilon\to 0$, $R\to\infty$) gives

$$I(s) = -I_-(s) - I_+(s),$$

$I_\pm$ the integrals over the vertical half-lines from $\pm 1$ upward, oriented upward for $I_+$ and downward for $I_-$ appropriately. On the contour of $I_+$, $z = 1+iy$:

$$I_+(s) = ie^{is}\int_0^\infty e^{-sy}(1-(1+iy)^2)^{\nu-1/2}\,dy, \tag{4}$$

and similarly for $I_-$. In (3), $e^{isx}$ oscillates rapidly; in (4), $e^{-sy}$ decreases rapidly except near $y = 0$, so the main contribution comes from $y$ near 0:

**Proposition 1.2**

Suppose $a$ and $m$ are fixed, with $a > 0$ and $m > -1$. Then as $s \to \infty$

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(e^{-cs}), \tag{5}$$

for some positive $c$.

**Proof.** $m > -1$ guarantees convergence at 0. Write $\int_0^a = \int_0^\infty - \int_a^\infty$. The first equals $s^{-m-1}\Gamma(m+1)$ (change variables $x \mapsto x/s$). For the second,

$$\int_a^\infty e^{-sx} x^m \, dx = e^{-cs} \int_a^\infty e^{-s(x-c)} x^m \, dx = O(e^{-cs}) \tag{6}$$

whenever $c < a$. ∎

**Proof of Theorem 1.1.** $(1-(1+iy)^2)^{\nu-1/2} = (-2iy)^{\nu-1/2} + O(y^{\nu+1/2})$ for $0 \le y \le 1$, and $= O(y^{\nu-1/2} + y^{2\nu-1})$ for $y \ge 1$. Proposition 1.2 with $a = 1$, $m = \nu - 1/2$ (and $\nu + 1/2$), together with (6), gives

$$I_+(s) = i(-2i)^{\nu-1/2} e^{is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2}), \qquad I_-(s) = i(2i)^{\nu-1/2} e^{-is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2})$$

(with the appropriate sign conventions). Since $J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)}[-I_-(s) - I_+(s)]$ and $\Gamma(1/2) = \sqrt{\pi}$, combining the two conjugate terms gives the theorem: $(-2i)^{\nu-1/2}$ and $(2i)^{\nu-1/2}$ contribute the phases $e^{\mp i\pi(\nu-1/2)/2}$, and the sum is $2^{\nu+1/2} s^{-\nu-1/2}\Gamma(\nu+\frac{1}{2})\cos(s - \frac{\pi\nu}{2} - \frac{\pi}{4})$ up to $O(s^{-\nu-3/2})$. ∎

Under restricted circumstances Proposition 1.2 extends to $\mathrm{Re}(s) \ge 0$:

**Proposition 1.3**

Suppose $a$ and $m$ are fixed, with $a > 0$ and $-1 < m < 0$. Then as $|s| \to \infty$ with $\mathrm{Re}(s) \ge 0$,

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(1/|s|).$$

(Here $s^{-m-1}$ is the branch that is positive for $s > 0$.)

**Proof.** First, for $\mathrm{Re}(s) \ge 0$, $s \ne 0$, $\int_0^\infty e^{-sx} x^m \, dx = \lim_{N\to\infty} \int_0^N e^{-sx} x^m \, dx$ exists and equals $s^{-m-1}\Gamma(m+1)$. Write $\int_0^N = \int_0^a + \int_a^N$; the first is entire in $s$. Since $-\frac{1}{s}\frac{d}{dx}(e^{-sx}) = e^{-sx}$, integration by parts gives

$$\int_a^N e^{-sx} x^m \, dx = -\frac{m}{s}\int_a^N e^{-sx} x^{m-1} \, dx - \Big[\frac{e^{-sx}x^m}{s}\Big]_a^N. \tag{7}$$

As $m < 0$, $\int_a^\infty x^{m-1} dx < \infty$, so letting $N \to \infty$, $\int_a^\infty e^{-sx} x^m \, dx$ is analytic in $\mathrm{Re}(s) > 0$ and continuous on $\mathrm{Re}(s) \ge 0$, $s \ne 0$; it equals $s^{-m-1}\Gamma(m+1) - \int_0^a$ for $s > 0$, hence everywhere there. Finally $\int_0^a = \int_0^\infty - \int_a^\infty$, and by (7), $\int_a^\infty e^{-sx} x^m \, dx = O(1/|s|)$. ∎

(Better error terms, or other ranges of $m$, require smooth cut-offs to mitigate the end-point $x = a$.)

## 2 Laplace's method; Stirling's formula

Consider $\int_a^b e^{-s\Phi(x)}\psi(x) \, dx$ with real $\Phi$, and $\Phi, \psi$ indefinitely differentiable. Assume there is $x_0 \in (a,b)$ with $\Phi'(x_0) = 0$ and $\Phi''(x) > 0$ throughout $[a,b]$.

**Proposition 2.1 (Laplace's method)**

Under the above assumptions, with $s > 0$ and $s \to \infty$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = e^{-s\Phi(x_0)}\Big[\frac{A}{s^{1/2}} + O\Big(\frac{1}{s}\Big)\Big], \qquad A = \frac{(2\pi)^{1/2}\psi(x_0)}{(\Phi''(x_0))^{1/2}}. \tag{8}$$

**Proof.** Replacing $\Phi$ by $\Phi - \Phi(x_0)$, assume $\Phi(x_0) = 0$. Since $\Phi'(x_0) = 0$, $\frac{\Phi(x)}{(x-x_0)^2} = \frac{\Phi''(x_0)}{2}\varphi(x)$ with $\varphi$ smooth, $\varphi(x) = 1 + O(x - x_0)$. Near $x_0$ change variables $y = (x - x_0)(\varphi(x))^{1/2}$; then $dx/dy = 1 + O(y)$ and $\psi(x) = \tilde{\psi}(y) = \psi(x_0) + O(y)$. On a small interval $[a', b'] \ni x_0$,

$$\int_{a'}^{b'} e^{-s\Phi(x)}\psi(x)\,dx = \psi(x_0)\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}\,dy + O\Big(\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}|y|\,dy\Big), \tag{9}$$

$\alpha < 0 < \beta$. With $y^2 = X$, $dy = \frac{1}{2}X^{-1/2}dX$, and (5), the first integral is $s^{-1/2}(\frac{2\pi}{\Phi''(x_0)})^{1/2} + O(e^{-\delta s})$; similarly the second is $O(1/s)$. On $[a, a']$ and $[b', b]$, $\Phi \geq c > 0$, so those integrals decay exponentially. ■

(8) extends to complex $s$ with $\mathrm{Re}(s) \geq 0$, using integration by parts to handle the oscillations of $e^{-s\Phi}$:

**Proposition 2.2**

With the same assumptions on $\Phi$ and $\psi$, the relation (8) continues to hold if $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$.

**Proof.** Proceed to (9); the first term is handled by Proposition 1.3 with $m = -1/2$. Observation: if $\Phi \geq 0$ and $|\Phi'(x)| \geq c > 0$ on $[a, b]$, then for $\mathrm{Re}(s) \geq 0$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = O\Big(\frac{1}{|s|}\Big) \qquad \text{as } |s| \to \infty; \tag{10}$$

indeed the integral is $-\frac{1}{s}\int_a^b \frac{d}{dx}(e^{-s\Phi})\frac{\psi}{\Phi'}\,dx = \frac{1}{s}\int_a^b e^{-s\Phi}\frac{d}{dx}\Big(\frac{\psi}{\Phi'}\Big)dx - \frac{1}{s}\Big[e^{-s\Phi}\frac{\psi}{\Phi'}\Big]_a^b$, and $|e^{-s\Phi}| \leq 1$. This handles $[a, a']$ and $[b', b]$, where $|\Phi'| \geq c > 0$ (as $\Phi'(x_0) = 0$ and $\Phi'' \geq c_1 > 0$). The second term of (9) has the form $\int e^{-s\frac{\Phi''(x_0)}{2}y^2}y\eta(y)\,dy$ with $\eta$ differentiable; write it as $\frac{-1}{s\Phi''(x_0)}\int \frac{d}{dy}(e^{-s\frac{\Phi''(x_0)}{2}y^2})\eta(y)\,dy$ and integrate by parts to get $O(1/|s|)$. ■

For purely imaginary $s = it$ this is the **method of stationary phase**; points with $\Phi'(x_0) = 0$ are **critical points**.

For $\delta > 0$ let $S_\delta = \{s : |\arg s| \leq \pi - \delta\}$, and let $\log s$ be the principal branch.

**Theorem 2.3 (Stirling's formula)**

If $|s| \to \infty$ with $s \in S_\delta$, then

$$\Gamma(s) = e^{s\log s}e^{-s}\frac{\sqrt{2\pi}}{s^{1/2}}\Big(1 + O\Big(\frac{1}{|s|^{1/2}}\Big)\Big). \tag{11}$$

(With more effort the error term improves to $O(1/|s|)$, with a full asymptotic expansion in powers of $1/s$. Note $\Gamma(s) \sim \sqrt{2\pi}s^{s-1/2}e^{-s}$, the usual form.)

**Proof.** First let $\mathrm{Re}(s) \geq 0$, with uniform error terms off a neighborhood of 0. For $s > 0$,

$$\Gamma(s) = \int_0^\infty e^{-x}x^s\frac{dx}{x} = \int_0^\infty e^{-x+s\log x}\frac{dx}{x} = e^{s\log s}e^{-s}\int_0^\infty e^{-s\Phi(x)}\frac{dx}{x}, \qquad \Phi(x) = x - 1 - \log x,$$

after $x \mapsto sx$; by analytic continuation $\Gamma(s) = e^{s\log s}e^{-s}I(s)$ for $\mathrm{Re}(s) > 0$, $I(s) = \int_0^\infty e^{-s\Phi(x)}\frac{dx}{x}$. It suffices to show

$$I(s) = \frac{\sqrt{2\pi}}{s^{1/2}} + O\Big(\frac{1}{|s|}\Big) \qquad \text{for } \mathrm{Re}(s) \geq 0. \tag{12}$$

$\Phi(1) = \Phi'(1) = 0$, $\Phi''(x) = 1/x^2 > 0$, $\Phi''(1) = 1$; so $\Phi$ is convex, positive, with minimum at 1. Apply Proposition 2.2 on $[1/2, 2]$ with $x_0 = 1$, $\psi(x) = 1/x$: this gives (12) there. On $[0, 1/2]$, since $\Phi'(x) = 1 - 1/x$,

$$\int_\epsilon^{1/2} e^{-s\Phi(x)}\frac{dx}{x} = -\frac{1}{s}\int_\epsilon^{1/2}\frac{d}{dx}(e^{-s\Phi(x)})\frac{dx}{\Phi'(x)x} = -\frac{1}{s}\Big[\frac{e^{-s\Phi(x)}}{x-1}\Big]_\epsilon^{1/2} - \frac{1}{s}\int_\epsilon^{1/2} e^{-s\Phi(x)}\frac{dx}{(x-1)^2}.$$

Since $\Phi(\epsilon) \to +\infty$ as $\epsilon \to 0$ and $|e^{-s\Phi}| \leq 1$, in the limit $\int_0^{1/2} e^{-s\Phi}\frac{dx}{x} = \frac{2}{s}e^{-s\Phi(1/2)} - \frac{1}{s}\int_0^{1/2} e^{-s\Phi}\frac{dx}{(x-1)^2} = O(1/|s|)$. $\int_2^\infty$ is treated analogously, since $\int_2^\infty (x-1)^{-2}dx < \infty$. So (11) holds for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

To pass to $\mathrm{Re}(s) \leq 0$, $s \in S_\delta$: for $\mathrm{Re}(s) \geq 0$, $s = \sigma + it$, $t \neq 0$, $\log(-s) = \log s - i\pi$ if $t > 0$ and $\log s + i\pi$ if $t < 0$. Hence for $G(s) = e^{s\log s}e^{-s}$,

$$G(-s)^{-1} = e^{s\log s}e^{-s}e^{-si\pi} \ (t > 0), \qquad G(-s)^{-1} = e^{s\log s}e^{-s}e^{si\pi} \ (t < 0). \tag{13}$$

Also

$$\Gamma(s)\Gamma(-s) = \frac{-\pi}{s\sin\pi s}, \tag{14}$$

from $\Gamma(s)\Gamma(1-s) = \pi/\sin\pi s$ and $\Gamma(1-s) = -s\Gamma(-s)$ (Theorem 1.4 of Chapter 6, Lemma 1.2 of Chapter 6). Combining (13), (14) and $(1 + O(|s|^{-1/2}))^{-1} = 1 + O(|s|^{-1/2})$ extends (11) to $S_\delta$ (in $S_\delta$, $\sin\pi s$ is comparable to $e^{\pm i\pi s}/(\mp 2i)$ for large $|t|$, and near the negative real axis $s$ is excluded). ■

## 3 The Airy function

The Airy function appeared in optics (intensity of light near a caustic):

$$\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(x^3/3+sx)}\,dx, \qquad s \in \mathbb{R}. \tag{15}$$

The integral converges and is continuous in $s$: $\frac{1}{i(x^2+s)}\frac{d}{dx}e^{i(x^3/3+sx)} = e^{i(x^3/3+sx)}$, so for $a \geq 2|s|^{1/2}$,

$$\int_a^R e^{i(x^3/3+sx)}\,dx = \int_a^R \frac{1}{i(x^2+s)}\frac{d}{dx}e^{i(x^3/3+sx)}\,dx, \tag{16}$$

and integrating by parts and letting $R \to \infty$ shows uniform convergence for $|s| \leq a^2/4$; similarly on $(-\infty, -a]$. Replacing $\mathbb{R}$ by $L_\delta = \{x + i\delta : x \in \mathbb{R}\}$, $\delta > 0$, improves matters: by Cauchy's theorem on rectangles $\gamma_R$ with $f(z) = e^{i(z^3/3+sz)}$, which is $O(e^{-\delta x^2})$ on $L_\delta$ and $O(e^{-yR^2})$ on the vertical sides,

$$\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{L_\delta} e^{i(z^3/3+sz)}\,dz,$$

and since $f(z) = O(e^{-\delta x^2})$ on $L_\delta$ for every complex $s$, Ai extends to an entire function. It satisfies

$$\mathrm{Ai}''(s) = s\,\mathrm{Ai}(s) : \tag{17}$$

$\mathrm{Ai}''(s) - s\,\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{L_\delta}(-z^2 - s)e^{i(z^3/3+sz)}dz = \frac{i}{2\pi}\int_{L_\delta}\frac{d}{dz}(f(z))\,dz = 0$, since $f$ vanishes at infinity along $L_\delta$. Comparing (17) with $y'' = Ay$ ($A$ large) suggests $\mathrm{Ai}(s)$ decreases rapidly as $s \to \infty$ (it stays bounded by (16), excluding $e^{\sqrt{A}s}$) and oscillates as $s \to -\infty$.

**Theorem 3.1**

Suppose $u > 0$. Then as $u \to \infty$, (i) $\mathrm{Ai}(-u) = \pi^{-1/2}u^{-1/4}\cos\left(\frac{2}{3}u^{3/2} - \frac{\pi}{4}\right)(1 + O(1/u^{3/4}))$.
(ii) $\mathrm{Ai}(u) = \dfrac{1}{2\pi^{1/2}}u^{-1/4}e^{-\frac{2}{3}u^{3/2}}(1 + O(1/u^{3/4}))$.

**Proof.** (i) Change variables $x \mapsto u^{1/2}x$ with $s = -u$: $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$, where

$$I_-(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{it(x^3/3-x)}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx \tag{19}$$

with $s = -it$, $\Phi(x) = x^3/3 - x$. $\Phi'(x) = x^2 - 1$: two critical points $x_0 = \pm 1$; $\Phi''(x) = 2x$, $\Phi(\pm 1) = \mp 2/3$. Split into $[-2, 0]$, $[0, 2]$ (one critical point each) and $(-\infty, -2]$, $[2, \infty)$. On $[0, 2]$, Proposition 2.2 with $s = -it$, $x_0 = 1$, $\psi = 1/2\pi$, $\Phi(1) = -2/3$, $\Phi''(1) = 2$ gives $\frac{1}{2\sqrt{\pi}}e^{-i\frac{2}{3}t}\left[\frac{1}{(-it)^{1/2}} + O(\frac{1}{|t|})\right]$; the interval $[-2, 0]$ gives $\frac{1}{2\sqrt{\pi}}e^{i\frac{2}{3}t}\left[\frac{1}{(it)^{1/2}} + O(\frac{1}{|t|})\right]$ (there $\Phi$ has a maximum; apply the proposition to $-\Phi$ with $s = it$). The complementary integrals are $O(1/|t|)$ by integration by parts ($\Phi' \neq 0$ there), e.g. $\int_{-\infty}^{-2} e^{it\Phi}dx = \lim_N \frac{1}{it}\int_{-N}^{-2}\frac{d}{dx}(e^{it\Phi})\frac{dx}{\Phi'(x)}$. Adding and inserting into $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$ gives (i).

(ii) With $s = u$, $\mathrm{Ai}(u) = u^{1/2}I_+(u^{3/2})$,

$$I_+(s) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx, \qquad F(x) = -i(x^3/3 + x). \tag{20}$$

There is no real critical point; repeated integration by parts shows rapid decay, but not its exact order. Follow the Bessel-function principle: deform the line to a contour $\gamma$ on which (a) $\mathrm{Im}(F) = 0$, and (b) $\mathrm{Re}(F)$ has a non-degenerate minimum at some $z_0$ (second derivative along $\gamma$ positive); then $F'(z_0) = 0$. Through such $z_0$ pass two orthogonal curves $\gamma_1$, $\gamma_2$ on which $F$ is real, $\mathrm{Re}(F)|_{\gamma_1}$ having a minimum and $\mathrm{Re}(F)|_{\gamma_2}$ a maximum at $z_0$; take $\gamma = \gamma_1$—the **method of steepest descent** ($-\mathrm{Re}\,F$ has a saddle at $z_0$). Here $\mathrm{Re}(F) = x^2y - y^3/3 + y$, $\mathrm{Im}(F) = -x^3/3 + xy^2 - x$, $F'(z) = -i(z^2 + 1)$, critical points $z_0 = \pm i$. At $z_0 = i$ the curves with $\mathrm{Im}(F) = 0$ are $\gamma_1 = \{y^2 = x^2/3 + 1\}$ and $\gamma_2 = \{x = 0\}$; on $\gamma_2$, $\mathrm{Re}(F)$ has a maximum at $i$, so take $\gamma = \gamma_1$, the branch $y = (x^2/3 + 1)^{1/2}$ of a hyperbola, asymptotic to the rays $re^{i\pi/6}$, $re^{i5\pi/6}$. Then

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx = \frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz, \tag{21}$$

by Cauchy's theorem on the region between $\mathbb{R}$ and $\gamma$ cut off by the circle of radius $R$ (there $e^{-sF(z)} = O(e^{-cyx^2})$, so the arcs contribute $O(\int_0^{\pi/6} e^{-cR^2\sin\theta}R\,d\theta) = O(1/R)$). On $\gamma$, $\Phi(x) = \mathrm{Re}(F) = y(x^2 - y^2/3 + 1) = (\frac{8}{9}x^2 + \frac{2}{3})(x^2/3 + 1)^{1/2}$, and $dz = dx + i\frac{x}{3}(x^2/3 + 1)^{-1/2}dx$; since $\Phi$ is even and $x(x^2/3 + 1)^{-1/2}$ odd,

$$\frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx. \tag{22}$$

As $(1 + u)^{1/2} = 1 + u/2 + O(u^2)$, $\Phi(x) = \frac{2}{3} + x^2 + O(x^4)$, so $\Phi(0) = 2/3$, $\Phi''(0) = 2$. Proposition 2.1 on $[-c, c]$ with $\psi = 1/2\pi$ gives $e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$; the rest is $O(e^{-2s/3}e^{-\delta s})$, since there $\Phi(x) \geq \frac{2}{3} + c_1x^2$. So $I_+(s) = e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$, which gives (ii). ∎

# 4 The partition function

**Theorem 4.1 (Hardy–Ramanujan)**

If $p$ denotes the partition function, then (i) $p(n) \sim \frac{1}{4\sqrt{3}\,n} e^{Kn^{1/2}}$ as $n \to \infty$, where $K = \pi\sqrt{2/3}$.
(ii) A much more precise assertion is that

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn}\Big(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\Big) + O(e^{\frac{K}{2}n^{1/2}}).$$

(Since $(n-\frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, $e^{K(n-1/24)^{1/2}} \sim e^{Kn^{1/2}}$, and (ii) implies (i).)

*Heuristic principle:* if $\sum F_n w^n$ has the unit disc as disc of convergence, the asymptotic behavior of $F_n$ is determined by the location and nature of the "singularities" of $F$ on the unit circle, each contributing according to its "order". (E.g. if $F$ is meromorphic in a larger disc with only a pole of order $r$ at $w = 1$ on the circle, then $F_n = P(n) + O(e^{-\epsilon n})$ for a polynomial $P$ of degree $r-1$, $\sum P(n)w^n$ being essentially the principal part.)

By Euler's formula (Theorem 2.1), with $w = e^{2\pi i z}$, $z = x + iy$, $y > 0$,

$$\sum_{n=0}^{\infty} p(n) e^{2\pi i n z} = f(z) = \prod_{n=1}^{\infty} \frac{1}{1 - e^{2\pi i n z}}, \qquad p(n) = \int_\gamma f(z) e^{-2\pi i n z}\, dz, \tag{23}$$

$\gamma$ the segment from $-1/2 + i\delta$ to $1/2 + i\delta$, $\delta > 0$ to be chosen in terms of $n$. $f$ is largest near $z = 0$: $|f(x+iy)| \le f(iy)$ (positive coefficients), increasing as $y$ decreases. Fortunately $f$ is a variant of the Dedekind eta function $\eta(z) = e^{\pi i z/12}\prod(1 - e^{2\pi i n z})$: $f(z) = e^{\pi i z/12}\eta(z)^{-1}$ (the $1/12$ explains the $1/24$ in the formula). By $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$ (Proposition 1.9),

$$f(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} f(-1/z), \tag{24}$$

and as $\mathrm{Im}(-1/z) \to \infty$, $f(-1/z) \to 1$ rapidly, since

$$f(z) = 1 + O(e^{-2\pi y}), \qquad z = x + iy,\ y \ge 1. \tag{25}$$

So $f_1(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}}$ approximates $f$ near 0, and by (24) $p(n) = p_1(n) + E(n)$ with

$$p_1(n) = \int_\gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz, \qquad E(n) = \int_\gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}(f(-1/z) - 1)\, dz.$$

*The error term.* For $z \in \gamma$, since $\mathrm{Re}(i/z) = \delta/(\delta^2 + x^2)$,

$$\Big|\sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\Big| \le c e^{2\pi n\delta} e^{\frac{\pi}{12}\frac{\delta}{\delta^2+x^2}}. \tag{26}$$

By (25) with $z$ replaced by $-1/z$,

$$|f(-1/z) - 1| \le c e^{-2\pi\frac{\delta}{\delta^2+x^2}} \qquad \text{if } \frac{\delta}{\delta^2+x^2} \ge 1; \tag{27}$$

and since $|f(z)| \le f(iy) \le C e^{\frac{\pi}{12y}}$ for $y \le 1$ (by (24)),

$$|f(-1/z) - 1| = O(e^{\frac{\pi}{12}\frac{\delta^2+x^2}{\delta}}) = O(e^{\frac{\pi}{48\delta}}) \qquad \text{if } \frac{\delta}{\delta^2+x^2} \le 1, \tag{28}$$

as $|x| \le 1/2$. Using (26), (27) when $\frac{\delta}{\delta^2+x^2} \ge 1$ gives $O(e^{2\pi n\delta})$ (as $2\pi > \pi/12$); using (26), (28) otherwise gives $O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$. Choose $\delta$ to minimize, $2\pi n\delta = \frac{\pi}{48\delta}$, i.e. $\delta = \frac{1}{4\sqrt{6}} n^{-1/2}$; then

$$E(n) = O(e^{\frac{\pi}{\sqrt{6}}n^{1/2}}) = O(e^{\frac{K}{2}n^{1/2}}).$$

*The main term.* Add to $\gamma$ the end-segments $[-1/2, -1/2+i\delta]$ and $[1/2+i\delta, 1/2]$, forming $\Gamma$; on them $\sqrt{z/i}\,e^{\frac{\pi i}{12z}} = O(1)$, so they contribute $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4}n^{1/2}})$, absorbed in $E(n)$. So

$$p_1(n) = \int_\Gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz. \tag{29}$$

Change variables $z \mapsto \mu z$ so that the exponentials combine to $e^{Ai(\frac{1}{z}-z)}$: this requires $A = 2\pi\mu(n-\frac{1}{24})$ and $A = \frac{\pi}{12\mu}$, i.e.

$$A = \frac{\pi}{\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{1/2}, \qquad \mu = \frac{1}{2\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{-1/2}.$$

Then

$$p_1(n) = \mu^{3/2}\int_{\Gamma'} e^{-sF(z)}\sqrt{z/i}\, dz, \qquad F(z) = i(z-1/z), \quad s = A, \tag{30}$$

$\Gamma' = \mu^{-1}\Gamma$ the union of $[-a_n, -a_n+i\delta']$, $[-a_n+i\delta', a_n+i\delta']$, $[a_n+i\delta', a_n]$, with $a_n = \frac{1}{2}\mu^{-1} \approx \sqrt{6}\, n^{1/2}$, $\delta' = \mu^{-1}\delta \to 1$. Steepest descent: $F$ has the critical point $z = i$ in $\mathbb{H}$; through it, $F$ is real on the imaginary axis (maximum, rejected) and on the unit circle (minimum at $i$). By Cauchy's theorem replace $\Gamma'$ by $\Gamma''$: the segments $[-a_n, -1]$, $[1, a_n]$ and the upper unit semicircle from $-1$ to 1. On the segments $F$ is purely imaginary, so they contribute $O(a_n^{3/2}\mu^{3/2}) = O(1)$. On the semicircle, $z = e^{i\theta}$ ($\theta$ from $\pi$ to 0), $dz = ie^{i\theta}d\theta$, $i(z-1/z) = -2\sin\theta$ and $\sqrt{z/i} = e^{i(\theta/2-\pi/4)}$; with $\theta = \varphi + \pi/2$ the integrand becomes $-e^{2s\cos\varphi}e^{3i\varphi/2}$, and the contribution is

$$\mu^{3/2}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\big(\cos(3\varphi/2) + i\sin(3\varphi/2)\big)\, d\varphi.$$

Laplace's method with $\Phi(\varphi) = -\cos\varphi$, $\varphi_0 = 0$, $\Phi(0) = -1$, $\Phi''(0) = 1$, $\psi(\varphi) = \cos(3\varphi/2) + i\sin(3\varphi/2)$, $\psi(0) = 1$, gives $\mu^{3/2}e^{2s}\frac{\sqrt{2\pi}}{(2s)^{1/2}}(1+O(s^{-1/2}))$. Since $2s = K(n-\frac{1}{24})^{1/2}$ and $\mu = \frac{1}{2\sqrt{6}}(n-\frac{1}{24})^{-1/2}$,

$$p(n) = \frac{1}{4\sqrt{3}\,n}e^{Kn^{1/2}}(1+O(n^{-1/4})),$$

which is (i).

*The precise formula (ii).* Write $p_1(n) = \frac{d}{dn}q(n) + e(n)$, where

$$q(n) = \frac{1}{2\pi}\int_\Gamma (z/i)^{-1/2} e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz$$

(differentiating $e^{-2\pi inz}$ in $n$ produces $-2\pi iz$, and $\frac{-2\pi iz}{2\pi}(z/i)^{-1/2} = (z/i)^{1/2}$), and $e(n)$ comes from the dependence of $\Gamma = \Gamma_n$ on $n$; by Cauchy's theorem it is $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4}n^{1/2}})$. As before,

$$q(n) = \frac{\mu^{1/2}}{2\pi}\int_{\Gamma''} e^{-sF(z)}(z/i)^{-1/2}\, dz, \tag{31}$$

the segments contributing $O(a_n^{1/2}\mu^{1/2}) = O(1)$ to $\frac{d}{dn}q(n)$. On the semicircle the integral equals

$$\frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\big(\cos(\varphi/2) + i\sin(\varphi/2)\big)\, d\varphi = \frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\cos(\varphi/2)\, d\varphi,$$

the odd part vanishing. With $\cos\varphi = 1 - 2\sin^2(\varphi/2)$ and $x = \sin(\varphi/2)$, $\cos(\varphi/2)d\varphi = 2dx$, this is

$$\frac{\mu^{1/2}}{\pi}e^{2s}\int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\, dx, \qquad \int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\, dx = \frac{\sqrt{\pi}}{2s^{1/2}} + O(e^{-2s}),$$

and the same holds after differentiating in $s$. Collecting the error terms,

$$p(n) = \frac{d}{dn}\Big(\frac{\mu^{1/2}e^{2s}}{2\sqrt{\pi}\,s^{1/2}}\Big) + O(e^{\frac{K}{2}n^{1/2}}).$$

Since $\mu^{1/2}/s^{1/2} = (2\pi)^{-1/2}(n-\frac{1}{24})^{-1/2}$ and $2s = K(n-\frac{1}{24})^{1/2}$, this is (ii).